**4.**

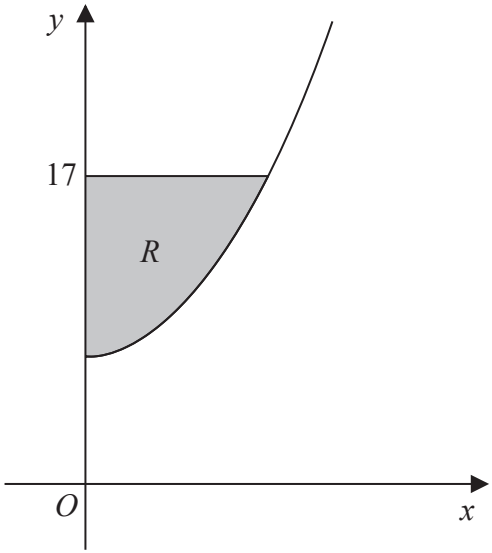



**Figure 1**

Figure 1 shows a sketch of the curve with equation

$$y = 2x^2 + 7 \qquad x \geqslant 0$$

The finite region $R$, shown shaded in Figure 1, is bounded by the curve, the $y$-axis and the line with equation $y = 17$

Find the exact area of $R$.

**(6)**